

PhiUSIIL Phishing URL Dataset

Feature Reference Guide — All 55 Features

PhiUSIIL: Phishing URLs with Integrated Syntactic, Semantic, and Infrastructure-based Lexical Features

Meta		3 features
FILENAME	Source file name the URL was collected from.	
URL	The raw URL string being analyzed.	
label	Target variable: 1 = phishing, 0 = legitimate.	

URL Structure		8 features
URLLength	Total number of characters in the URL. Phishing URLs are often longer to hide the true destination.	
Domain	Extracted domain portion of the URL (e.g. google.com).	
DomainLength	Number of characters in the domain name. Phishing sites often use longer or randomly generated domains.	
IsDomainIP	Binary flag: 1 if the domain is a raw IP address (e.g. 192.168.1.1) instead of a name — a red flag for phishing.	
TLD	Top-level domain extracted from the URL (e.g. .com, .net, .xyz).	
TLDLength	Length of the top-level domain string.	
NoOfSubDomain	Number of subdomains present. Phishing URLs often use multiple subdomains (e.g. login.bank.evil.com).	
IsHTTPS	1 if the URL uses HTTPS. Note: phishing sites increasingly use HTTPS, so this alone is not conclusive.	

URL Lexical		13 features
URLSimilarityIndex	Score measuring how lexically similar the URL is to known legitimate URLs. Higher similarity may indicate spoofing/typosquatting.	
CharContinuationRate	Rate of consecutive repeated characters in the URL. Obfuscated or padded URLs tend to have unnatural character runs.	
TLDLegitimateProb	Probability score for how commonly this TLD appears in legitimate websites. Derived from corpus statistics (e.g. .com ~ 0.92, .tk ~ 0.04).	
URLCharProb	Probability score based on character n-gram frequency in the URL. Gibberish/auto-generated domains have low scores.	
NoOfLettersInURL	Count of alphabetic characters in the full URL.	
LetterRatioInURL	Proportion of the URL that is alphabetic characters.	
NoOfDegitsInURL	Count of numeric digits in the URL.	
DigitRatioInURL	Proportion of the URL that consists of digits. High digit ratios can indicate IP-based or auto-generated URLs.	

NoOfEqualsInURL	Number of '=' characters. Many equals signs suggest complex query strings, sometimes used to hide phishing parameters.
NoOfQMarkInURL	Number of '?' characters in the URL (query string separators).
NoOfAmpersandInURL	Number of '&' characters, indicating multiple query parameters.
NoOfOtherSpecialCharsInURL	Count of special characters beyond standard alphanumeric ones (e.g. @, %, #). Phishing URLs often abuse special chars.
SpacialCharRatioInURL	Ratio of special characters to total URL length.

Obfuscation 3 features	
HasObfuscation	Binary flag: 1 if URL encoding tricks or character obfuscation is detected in the URL.
NoOfObfuscatedChar	Number of obfuscated/encoded characters found (e.g. %20, %2F, hex-encoded characters).
ObfuscationRatio	Ratio of obfuscated characters to total URL length.

Page Content 13 features	
LineOfCode	Number of lines in the page's HTML source code.
LargestLineLength	Length of the longest line in the HTML source. Very long lines can indicate minified or obfuscated scripts.
HasTitle	Binary flag: 1 if the page has an HTML tag.
Title	The actual text content of the page's tag.
DomainTitleMatchScore	How closely the page title matches the domain name. Low score on a 'bank login' page can indicate spoofing.
URLTitleMatchScore	Similarity score between the full URL and the page title. Mismatch is a phishing indicator.
HasFavicon	Binary flag: 1 if the page has a favicon. Phishing pages sometimes skip this detail.
Robots	1 if the site has a robots.txt file — legitimate sites usually do; many phishing sites do not.
IsResponsive	1 if the page appears to be mobile-responsive. Hastily built phishing pages often are not.
HasDescription	1 if the page has an HTML meta description tag.
NoOfImage	Total number of images embedded in the page.
NoOfCSS	Number of external CSS stylesheet links loaded.
NoOfJS	Number of external JavaScript files loaded. Many phishing pages load minimal scripts.

Behavior 9 features	
NoOfURLRedirect	Number of URL redirects detected. Multiple redirects are a common phishing tactic to obscure the final destination.
NoOfSelfRedirect	Number of redirects pointing back to the same domain.
NoOfPopup	Number of popup windows triggered by the page. Phishing pages sometimes use popups for credential harvesting.

NoOfiFrame	Number of iFrames on the page. Hidden iFrames are frequently used in phishing to load content invisibly.
HasExternalFormSubmit	1 if any form on the page submits data to an external (different) domain — a major phishing red flag.
HasSubmitButton	1 if the page contains a form submit button (login/credential submission indicator).
HasHiddenFields	1 if the page contains hidden form fields, which can be used to pass extra stolen data silently.
HasPasswordField	1 if the page contains a password input field — a strong signal when combined with other phishing indicators.
HasSocialNet	1 if the page contains links to social network sites (sometimes used to add legitimacy to phishing pages).

Keyword		4 features
Bank	1 if the URL or page content contains banking-related keywords (e.g. 'bank', 'account', 'secure').	
Pay	1 if payment-related keywords are present (e.g. 'pay', 'payment', 'checkout').	
Crypto	1 if cryptocurrency-related terms appear, targeting crypto wallet phishing.	
HasCopyrightInfo	1 if the page contains copyright notices. Legitimate sites often have them; phishing clones may forget to include them.	

References		3 features
NoOfSelfRef	Number of links pointing to the same domain (internal links). Legitimate sites usually have many; phishing pages often have few.	
NoOfEmptyRef	Number of href='#' or empty link references. Cloned phishing pages often have broken/empty links.	
NoOfExternalRef	Number of outbound links to other domains.	

Source: *PhiUSIIL Phishing URL Dataset* — UCI Machine Learning Repository